AI Security Applied to AIxBio
Policy Limitations in Dual-Use Risk Research and Technology Applications

Victoria Macali

Abstract

Emerging technologies bring about emerging regulation concerns, in this case, the security risks posed by AIxBio research resulting in dual use data. The data produced by AIxBio problems; prescription drug development, protein and genomics structure prediction, and overall biology optimization leaves data on the table at risk of positive and negative manipulation. As a Scientist of any discipline, ethics play a big role in the safeguarding and overall disclosure of AIxBio research. While the field itself is growing, so do the policy limitations once understanding that a strict and static approach will not suffice in this quickly evolving field. Regulation should however take an iterative approach to modeling and red teaming against its own data.

AI Security Risks in AIxBio

Aside from the concerning threat of day to day cybersecurity attack vectors, AIxBio brings about more insecurities researchers manipulate and distribute their data. Major concerns are raised by AI models for biological design, data sharing, confidentiality, and testing. While access to information at the foundational core to research and science, bad actors can gain highly sensitive information through the lighthearted decisions of researchers. The regulation proposal I suggest is to encourage constant interaction with the data and red teaming against predictive models to find insecurities that may be exploited.

Research Direction

AI Security and AIxBio share a common theme: industry standards and policy take longer to adopt and enforce against an ever-changing landscape. The Dual-use idea stems from how the discoveries made can be beneficial to United States civilians for helpful purposes while also can harm the quality of life of civilians from military, surveillance, and/or malicious acts. Building security tests and monitoring enable policy makers to identify low risk and what constitutes an update to high risk. Red team exercises can be seamlessly integrated into best practices, to support a risk adaptive model for responsible AI deployment.

Conclusion

The intersection of AI Security and AIxBio depicts the critical risks and testing needed to uphold national security and innovation. Policymaking should not be seen as an afterthought, but rather a support system built over time that can protect itself. Policy making can be adaptable and with proper procedures overseen by SME's can this field may grow into a totally new space.The goal of my research within the RAND CAST Fellowship concerns strictly about how systems are deployed throughout AI research and how to regulate a decentralized culture.